SERENDiB COLLEGE

Science – Grade 10

Assignment – 01

Deadline: 14.02.2026 | 11:59 PM

Read the below article carefully and answer the following questions in your writing book.

01. Read the article carefully and construct an appropriate table to summarise the different properties of water and explain how each property contributes to the maintenance of life.

Specific Property	Contribution to the Maintenance of Life

02. Based on the article, summarise the importance of minerals in maintaining human life using the table below.

Element	Functions	Deficiency symptoms

03. Construct a suitable table to summarise the minerals required by plants, explaining their role in plant growth and the visible symptoms that appear when they are deficient.

Element	Functions	Deficiency symptoms

04. Construct an appropriate table to summarise the different types of vitamins, their functions in the human body and the diseases or conditions caused by their deficiency.

Vitamin	Uses of the Vitamin	Deficiency symptoms

Note: In addition to completing the tables, attach at least four relevant pictures showing deficiency symptoms related to minerals and vitamins in humans and plants. Write a brief caption below each picture.

The Invisible Support System of Life: Water, Minerals and Vitamins

Life depends on essential substances that work continuously inside living organisms. Among them, water, minerals and vitamins play a major role in maintaining normal body functions in humans and plants. When present in correct amounts, they support growth and health. When deficient, clear symptoms appear.

The Role of Water in Maintaining Life

Water is an important inorganic compound in living organisms. It acts as a good solvent, allowing many substances to dissolve in it. Because of this, water provides a medium for biochemical reactions inside cells. It forms the main component of extracellular fluids and helps remove excretory materials from the body.

Water also acts as a respiratory medium. Oxygen dissolves in water, allowing aquatic organisms to obtain oxygen for respiration.

Another key property of water is its high specific heat capacity. This helps regulate body temperature, preventing sudden temperature changes despite environmental fluctuations.

Water functions as a transport medium as well. It forms the main component of blood and carries nutrients, vitamins and hormones throughout the body. In plants, cohesive and adhesive forces help water move upward to different parts of the plant.

In addition, water provides a living environment. Since ice is less dense than water, it floats on the surface, keeping the water below in liquid form. This allows aquatic organisms to survive in cold conditions.

Importance of Minerals in Maintaining Human Life

Minerals are inorganic nutrients required in different amounts. Some are needed in large quantities (macro elements) and others in small amounts (trace elements). Even though they form a small percentage of body weight, they are essential for normal growth and metabolism. When minerals are deficient, specific symptoms appear.

Potassium maintains ionic balance in cells and supports heart and muscle activity. It also helps transmit nerve impulses. Its deficiency causes muscle weakness and psychological disorders.

Sodium activates enzymes, forms part of digestive juices and maintains osmotic pressure in cells. It also assists nerve impulse transmission. Deficiency may lead to respiratory disorders, cramps, nausea and diarrhea.

Magnesium contributes to bones and teeth and regulates nerve activity in skeletal muscles. It supports metabolic processes. A lack of magnesium may cause high heartbeat and nerve irritability.

Calcium is essential for the growth of bones and teeth. It supports blood clotting, nerve function, milk production and absorption of Vitamin B. Deficiency results in rickets, weakened bones and growth disorders.

Phosphorous supports bone growth, forms part of nucleic acids and helps in carbohydrate and fat metabolism. It also enables quick energy release in muscles and nerves. Deficiency causes bones to become weak and fragile.

Iron is required for haemoglobin synthesis and oxygen storage in muscles. It is also part of certain enzymes. Iron deficiency leads to anaemia, sleepiness and reduced psychological development.

Iodine is needed for the production of the hormone thyroxin. Deficiency affects intelligence development, causes lethargy and may limit body height.

Importance of Minerals in Maintaining Plant Life

Plants require minerals from the soil for proper growth and development. Each mineral has a specific function, and deficiency produces visible symptoms.

Nitrogen forms part of amino acids, proteins, nucleic acids and chlorophyll. Its deficiency causes retarded growth and chlorosis in leaves.

Phosphorous is a component of nucleic acids and ATP and is important for root development. Deficiency leads to poor root growth and red or purple patches on leaves.

Potassium supports protein synthesis and controls the opening and closing of stomata. Deficiency causes chlorosis and yellow or brown patches on leaves.

Sulphur forms part of amino acids and proteins. When deficient, chlorosis appears in veins and between veins.

Iron is needed for chlorophyll and respiratory enzyme synthesis. Its deficiency causes chlorosis in young leaves.

Calcium strengthens the cell wall, maintains plasma membrane structure and supports enzyme activity. Deficiency leads to death of tissues and leaf tips.

Zinc is required for enzyme activity and chlorophyll synthesis. Deficiency results in dead tissues and thickened leaves.

Importance of Vitamins in the Human Body

Vitamins are organic compounds required in small amounts to regulate body processes. They are classified as water-soluble (Vitamin B and C) and fat-soluble (Vitamins A, D, E and K). Each vitamin performs specific functions, and deficiency leads to characteristic conditions.

Vitamin A supports vision and maintains healthy skin. Its deficiency causes night blindness, Bitot patches, dry skin and respiratory tract diseases.

Vitamin B maintains nerve function, healthy skin, bone marrow formation, red blood cell maturation and antibody production. Deficiency may cause beri beri, anaemia and reduced immunity.

Vitamin C supports skin health, enamel formation and collagen synthesis. Its deficiency leads to weak gums, internal bleeding and scurvy.

Vitamin D controls calcium and phosphorous absorption and supports bone health. Deficiency may result in tooth decay and osteoporosis.

Vitamin E supports tissue growth and cell protection. Deficiency may cause premature births and breakdown of red blood cells.

Vitamin K is required for blood clotting. Deficiency delays clot formation.

Water, minerals and vitamins are essential for maintaining life in humans and plants. Water supports biochemical reactions, transport and temperature regulation. Minerals contribute to structural strength and metabolic activities, while vitamins regulate body functions and prevent diseases. Adequate intake and balance of these substances ensure proper growth, health and survival.
